

Problem Set 8

Throughout this problem set, use Black's formula to price up swaptions and assume flat skew.

For problems 1–5 below, suppose that today's swap curve is as follows (these rates are annual bond and assume that the floating side is 1-year LIBOR paid annual bond as well):

Maturity	Swap Rate
1	5.25%
2	5.50%
3	5.75%
4	6.00%
5	6.25%

1. a) What is the carry and roll-down for a 2-year, 3-year, 4-year, and 5-year swap, for a horizon of one year? Specify your answer in terms of both bps upfront and bps running.

b) Suppose an investor wants to receive fixed, taking an outright view that the market will rally. Which of the swaps in part a) provides the best carry and roll-down?

c) Suppose an investor wants to pay fixed, taking an outright view that the market will sell off. Which of the swaps in part a) provides the best carry and roll-down?

d) Suppose an investor receives fixed on \$100mm of the swap you selected in part b). If in a year swap rates are unchanged from today, what is the P&L on the trade in dollars? If in a year swap rates have sold off by an amount equal to the carry and roll-down (expressed in bps running), what is the P&L on the trade in dollars?

2. A swap is referred to as being *positive carry* for a counterparty if the net cash flow to be received by the counterparty in the current accrual period in the swap is positive. Obviously, if a swap is positive carry to one counterparty in the swap, then it must be *negative carry* to the other counterparty.

a) Suppose that you wish to enter into a 5-year swap. If you wish to enter into a positive carry trade, will you pay fixed or receive fixed?

b) Suppose that you enter into the positive carry trade as you specified in a) above. Suppose that you intend to employ the following strategy: (1) enter into the positive carry swap, earning positive net cash flow in the first year, and (2) unwind the swap at the end of the first year after earning the positive carry. How much money should you expect to make employing this strategy? Justify your answer.

3. Consider the following three-legged strategy:

- Buy \$100mm 1 into 4 receivers struck at-the-money forward.
- Sell \$200mm 1 into 4 receivers struck *at-the-money spot*.⁶³
- Buy \$100mm 1 into 4 receivers swaptions struck at $x\%$.

a) Suppose that all 1 into 4 receivers swaptions are priced using implied vols of 20%. What strike of x gives this trade zero value today?

b) Graph the payout profile of this trade. What view are you taking if you put on this trade and intend to hold the position until expiration?

c) Suppose that in one year the swap curve is as follows:

Maturity	Swap Rate
1	5.00%
2	5.25%
3	5.50%
4	5.75%
5	6.00%

What is the value of the trade?

⁶³A swaption is said to be struck *at-the-money spot* if the strike is equal to the current swap rate for the underlying maturity.

4. Suppose that you pay fixed on \$100mm 4-year swaps and simultaneously receive fixed in 1-year swaps.

a) What notional of 1-year swaps should you do if you want to have the value of a bp running in both trades be the same? That is, what notional of 1-year swaps should you do if you want this trade to be duration neutral? If you do this trade, what view are you taking?

b) Suppose now that you pay fixed on \$100mm of a 4-year swap, one year forward and simultaneously receive fixed in a 1-year swap, one year forward. What notional of the 1x2 should you do to make this trade duration neutral? What are the two rates in the swap? If you do this trade, what view are you taking? How does this trade compare to the trade in part a), and is the entry level better or worse for you?

c) Suppose now that 1 into 4 year payer swaptions trade at implied vols of 20% and that 1 into 1 payer swaptions trade at implied vols of 25%. Suppose that you buy \$100mm of a 1 into 4 year payer swaption struck at-the-money forward and that you simultaneously sell 1 into 1 payer swaptions on the notional amount you solved for in part b). What strike on the 1 into 1 payer swaption do you solve for that gives this trade zero upfront cost? What view are you taking in this trade? How does your entry level compare to the entry level you found in part b)? Can you explain why the entry level is better or worse?

5. Suppose you purchase a 1 into 4 receiver swaption struck at-the-money forward.

a) How much does this cost, assuming 1 into 4 swaption vol is 20%?

b) Suppose that the swap curve moves only in parallel shifts. Assuming that you hold this position until expiry, where must 4-year swap rates be in order to break even on this trade? How does this break-even compare with the at-the-money forward strike? How does this break-even level compare with 4-year spot swap rates?

c) Suppose you purchase a 1 into 4 payer swaption struck at-the-money forward. How much does this cost, assuming 1 into 4 swaption vol is 20%?

d) Assuming that you hold this position until expiry, where do you need 4-year swap rates to be in order to break even on this trade? How does this

break-even level compare with the at-the-money forward strike? How does this break-even level compare with 4-year swap rates?

6. In Section 8.2.2, we discussed that a forward curve trade could be decomposed into four spot starting swaps. Consider again the swap curve in Table 8.1 and a 1s3s curve steepener, two years forward. As we discussed, putting this trade on and risking \$100,000 per bp amounts to receiving on \$1,167.98mm of a 2x3 swap at a rate of 5.712% and paying on \$412.57mm of a 2x5 swap at a rate of 5.976%.

a) Identify the four spot starting swaps in the decomposition of this forward curve trade, identifying maturities, directions, notionals, and coupons.

Referring to this decomposition, answer the following questions:

b) What is the present value of the carry over the first two years?

c) After the first two years, describe the remaining position. How does the rate differential in the two swaps that remain after two years compare to the rate differential in the forward curve trade?

d) Suppose that instead of earning carry over the first two years, we would like to forgo carry and adjust the rates of the two swaps in the decomposition that have a notional of \$1,167.98mm. The rates in these two swaps will be adjusted by the same amount so as to offset the present value of carry that occurs over the first two years. What is the magnitude of this adjustment? Given this adjustment, what is the rate differential that exists in the remaining position after two years? How does this adjusted rate differential compare to the rate differential in the forward curve trade? Explain.

e) For this question, forget part d) above. Suppose that instead of earning carry over the first two years, we would like to forgo carry and adjust the rates of the two swaps in the decomposition that have a notional of \$412.57mm. The rates in these two swaps will be adjusted by the same amount so as to offset the present value of carry that occurs over the first two years. What is the magnitude of the adjustment? Given this adjustment, what is the rate differential that exists in the remaining position after two years? How does this compare to our findings in part d)? Explain.

7. Consider the bearish steepener that we discussed in Section 8.2.3. In particular, consider the version of the trade with no upfront: we buy \$412.57mm

of a 2 into 3 payer swaption struck at 5.889%, and we sell \$1,167.98mm of a 2 into 1 payer swaption struck at 5.712%. Recall that the strike differential on this trade is $5.889\% - 5.712\% = 17.7$ bps.

Suppose that at the time at which we enter this trade we establish a \$150,000 stop-loss on the trade in a year: if the mark-to-market on the trade in one year is more than \$150,000 against us, we will unwind the trade so as to prevent any further loss.

Suppose now that one year has gone by and the swap curve is as follows:

Maturity	Swap Rate
1	5.00%
2	5.10%
3	5.20%
4	5.30%
5	5.40%

And suppose further that swaption vols in one year are as follows:

Swaption	Lognormal Vol
2 into 1	18%
2 into 3	16%
1 into 1	18%
1 into 3	16%

a) How does the strike differential in the trade compare to the forwards prevailing now? Is the forward differential prevailing now favorable compared to the strike differential?

b) Have we hit our stop-loss?

c) How do you reconcile your answers to a) and b)?

8. Consider our stylized example and the swap curve in Table 8.1.

a) Consider now a 3-year agency bond that has a coupon of 5% and a yield of $T_3 + 18$. What is the asset swap spread of this bond, assuming a par asset swap?

b) Consider again the 3-year agency bond that has a coupon of 5% and a yield of $T_3 + 18$. What is the asset swap spread of this bond, assuming a yield-yield asset swap?

c) Refer now to part b). Suppose that an investor buys the bond and enters into a yield-yield asset swap (i.e., the investor buys the bond and pays fixed in order to enter into a yield-yield asset swap). Suppose further that later in the day (after entering into this yield-yield asset swap) 3-year Treasury yields increase 5 bps, and this is the only change in the market. What is the mark-to-market of the asset swap?

d) For this question, forget part c) above. Suppose again that an investor buys the bond and enters into a yield-yield asset swap. Suppose further that later in the day (after entering into the yield-yield asset swap) 3-year swap spreads increase 2 bps, and the spread on the agency bond decreases from 18 bps to 17 bps. Assume that these are the only changes in the market. What is the mark-to-market of the asset swap?

e) Consider now a 3-year agency bond that has a coupon of 11.5% and a yield of $T_3 + 18$. What is the asset swap spread of this bond, assuming a par asset swap?

f) Consider again the 3-year agency bond that has a coupon of 11.5% and a yield of $T_3 + 18$. What is the asset swap spread of this bond, assuming a yield-yield asset swap?

g) How do your answers to part a) and part e) compare? How do your answers to part b) and part f) compare? Explain.

Chapter 9

More Recent Product Innovations

The few years preceding the financial crisis were marked by a number of new, exotic trades in the swaps market following what had been considered by many to be a long period of relatively little innovation. Some of these trades were somewhat pathbreaking in nature, whereas others simply borrowed well-known concepts from other markets and recast them in the swaps market. Still others were a by-product of other products whose popularity was on the rise.

The trades that we will look at in this chapter exhibit *correlation risk*—the feature that more than one underlying variable determines the value of the trade and the correlation between the variables must be taken into account to properly value the trade. A *correlation trade* is a trade that has correlation risk. Correlation trades have been around in certain markets for a long time, but the rates markets have not traditionally been thought of as a hotbed for them.¹

¹The credit markets come to many people's minds as a place where correlation trades abound. In a synthetic collateralized debt obligation (CDO), for example, a number of credit default swaps are placed in a "basket." Suppose that 100 credit default swaps are placed in a basket, and two tranches are formed: an equity tranche and a senior tranche. The senior tranche is protected against the first ten defaults in the portfolio, with the equity tranche absorbing those losses. What does an increase in pairwise correlation among defaults in the 100 names in the basket do to the value of each of the tranches? An increase in correlation decreases the value of the senior tranche and increases the value of the equity tranche. With 100 names in the basket, the senior tranche can withstand a few random blowups in the portfolio. However, as correlation increases, the occurrence of a single blowup implies that many other defaults are more likely to occur, which will erode the

That is not to say that the rates markets have been completely devoid of correlation trades. There was a trade that we looked at in Chapter 5 that is actually a correlation trade, and we'll take a look at it again shortly. But examples such as these correlation trades were traditionally few and far between in the rates markets. Then in 2004 to 2005, we started seeing a surge of correlation trades. One reason for the increase in these types of trades was the popularity of a certain type of structured note known as a non-inversion note, which has embedded correlation risk. As we will discuss, as more and more of these structured notes were printed, dealers started getting long a lot of options with correlation risk. In order to offload or offset some of this risk in their books, it became desirable for dealers to come up with trades that had embedded correlation risk and could be marketed to investors. This led to the birth of curve options.

In this chapter we will focus on two products: forward vol and curve options. Although these two trades are quite distinct, they have some similarities: their popularity grew as an offshoot of other products, and, as mentioned, they are both examples of correlation trades. It should be noted that the pricing of these products is complex and is outside the scope of this text. Nonetheless, we will be able to make some progress in describing the issues and challenges involved in accurately pricing these complicated products.

9.1 An Introduction to Correlation Trades: Caps Versus Payer Redux

In Section 5.2, we talked about how caps and payer swaptions are closely related. Look again for a moment at Figure 5.4, which is a representation of a 2x5 cap on 3-month LIBOR and a 2 into 3 payer swaption (with a quarterly money strike versus 3-month LIBOR). We discussed some similarities and differences between these trades and, in particular, how the 2x5 cap is a portfolio of options on forward rates, and the 2 into 3 payer swaption is an option on a portfolio of forward rates. What we did not discuss at the time is that another difference between the two trades can be explained by correlation.

If the strikes on the two options are the same, then we said that the cap must be worth more than the payer swaption. But how much more? The answer depends on correlation. The two trades are written on the exact same set of 12 3-month LIBOR forwards. The degree of pairwise correlation

value of the senior tranche. Thus, as correlation goes up, the senior tranche is hurt. Inasmuch as the risk of the entire basket is unaffected by an increase in correlation, this must mean that as correlation goes up, the value of the equity tranche must also go up.

between forward rates determines the relative pricing of the cap and the payer swaption.

To see that correlation plays a role in the relative value of the cap versus the payer swaption, let's consider a couple of cases. First, suppose that each of the forward rates is pairwise uncorrelated. In this case, if the forward rates are volatile, then there will be a lot of value to the cap, which is a portfolio of options on the individual forward rates. However, even though there may be a lot of movement in the individual forward rates, the 3-year swap rate, two years forward will not be too volatile in this case; as some of the individual LIBOR rates go up, others may go down, canceling one another out and leaving the forward swap rate relatively stable. In this case, the cap will be much more valuable than the payer swaption.

But suppose instead that the 12 forward rates are all pairwise highly correlated. Then as one forward rate goes up, they will all tend to go up, and that implies that the 3-year swap rate, two years forward will tend to go up as well. Similarly, if a 3-month LIBOR forward goes down, then they will all tend to go down, in which case the forward swap rate will tend to go down as well. In this case, a lot of volatility in the LIBOR forwards tends to mean a lot of volatility in the 3-year swap rate, two years forward. Thus, in the case of high pairwise correlation between each of the forward LIBOR rates, the cap will still be worth more than the payer swaption, but the differential in price will not be as extreme as the case in which all of the forward LIBOR rates are pairwise uncorrelated.²

Caps versus payers has been a popular relative value correlation trade for years. It refers to a trade in which an investor buys a cap and sells a payer swaption (or vice versa), with a view that a move in correlation will make the long increase in value relative to the short. For example, if an investor expects that the pairwise correlation between forward LIBOR rates will increase in the near future, he may want to put on a trade in which he sells caps and buys a payer swaption today. If the pairwise correlation between forward LIBOR rates does indeed increase, the trade may subsequently be unwound at a profit.

9.2 Forward Vol Trades

A forward vol trade is a rather elegant product that became mainstream around 2004. It is, as the name suggests, a forward contract on (implied)

²This is an application of results presented in Merton (1973). He shows that the value of an option on a portfolio must be less than or equal to the value of a corresponding portfolio of options.

volatility.³ A forward vol trade therefore allows an investor to express a view on the direction of implied volatility. Prior to the creation of this product, an investor who was interested in expressing a view on implied volatility would most likely trade some options. For example, an investor who thought that implied 5 into 10 swaption vol would go up might buy some 5 into 10 swaption straddles struck at-the-money forward.⁴ However, with the passage of time and the movement of rates, the buyer of a swaption straddle has a position that is affected by much more than just movements of 5 into 10 swaption vol. For instance, the straddle position may acquire significant delta exposure as rates move and the strike is no longer at-the-money forward.⁵ Moreover, with the passage of time, the buyer of a 5 into 10 swaption straddle will no longer have exposure to 5 into 10 implied vol. For example, after a year, the buyer will have exposure to 4 into 10 vol. Someone who enters a forward vol trade, by comparison, avoids many of the issues faced when trading a swaption straddle in order to take a view on implied vol.

9.2.1 Preliminary

Before diving in and tackling forward vol, let's take a look at the following (seemingly unrelated) trade. Consider the *swaption on a forward starting swap* depicted in Figure 9.1. This is a 2 into 5x15 receiver swaption (which is pronounced "two into five by fifteen").⁶

³Prior to 2004, forward vol was a concept that was implicit in the valuation of Bermudan swaptions; as we alluded to in Chapters 5 and 6 and will discuss further later in this chapter, the decision to exercise a Bermudan swaption on an expiration date depends in part on expectations of what implied volatility will be in the future. But it was not until around 2004 that the concept of trading forward vol as a stand-alone product emerged.

⁴An investor who wanted to express the view that 5 into 10 implied vol would go up could alternatively buy either a payer swaption or a receiver swaption. However, buying one of these options in isolation may also give the investor substantial delta risk. By contrast, a swaption straddle struck at-the-money forward will be delta-neutral or nearly delta-neutral at trade inception, depending on what model we use (question 12 of Problem Set 5 gave you an opportunity to ponder this).

⁵Very often, investors buy swaption straddles struck at-the-money forward to take a view on vol. Such straddles have limited delta initially but may acquire significant delta as rates move around. Oftentimes, investors will use swaps to hedge any acquired delta position in order to maintain a position that has exposure to vol, not an exposure to rates.

⁶There is a lot of jargon being employed in the description of this trade, so let's review a few things. The terminology " x into y " for swaptions refers to an option that expires in x years and gives one the right to enter into a y -year swap. Regarding forward starting swaps, the terminology " $x \times y$ " refers to a swap that starts x years from today and ends y years from today (and thus the swap lasts $y - x$ years). Thus, a 2 into 5x15 receiver swaption is a receiver swaption that gives the owner the right in two years to receive in a 10-year swap, five years forward.